



Juan Fernando Garcia Navas

• Seeking an industrial internship · from July 2027

CONTACT

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Nancy, France **Open to relocation**

LANGUAGES

Spanish **Native**

English **C1**

French **B2**

SOFTWARE

Aspen HYSYS / PRO II ★★★★★

Aspen Plus ★★★★★

Python / MATLAB / Excel ★★★★★

Fusion 360 ★★★★★

Claude / Claude Code ★★★★★

SCAN

The full portfolio



juanfermandogarcia.github.io/portfolio

 **ENSIC** Degree Double degree · ENSIC + Master + UIS

Programme Chemical and process engineering · ENSIC 3rd year + Master

Specialisation Processes for Energy and the Environment

Profile

Double degree between ENSIC and UIS. Drawn to the energy transition: CO₂ valorisation, hydrogen, effluent treatment and energy reduction, with projects taken from simulation to real assembly.

Skills

- Mass and energy balances
- Risk analysis (MOSAR, HAZOP, P&ID)
- Experimental work and project management
- Equipment sizing
- Techno-economic assessment

Experience

- JAN · JUN 2027 **Research internship · LRGP, ENSIC** 
Modelling and simulation of a membrane contactor for CO₂ capture with potassium carbonate solutions.
- ONGOING **Innovation project · Alfa Laval** 
Innovative valorisation of brazed aluminium heat exchangers rejected for defects.
- 2026 **Ammonia unit · Technip Energies** 
Full design of a 2 700 t/day unit, defended before their engineers.
- 2025 **Cooling tower · UIS**
From technology study to live start-up, 73.1% thermal efficiency.
- 2025 **Research intern · CICAT, UIS**
Synthesis and characterisation of nickel catalysts.

Education

- 2025 · PRESENT
Diplôme d'ingénieur + Master · Chemical and Process Engineering
ENSIC, Nancy, France. 3rd year. Specialisation: Processes for Energy and the Environment.
- 2020 · PRESENT
Chemical Engineering
Universidad Industrial de Santander, Colombia. 5th year.

Publications and posters

Poster · Ni/TiO₂/Al₂O₃ catalysts for CO₂ methanation
Colombian National Symposium on Catalysis, 2025 · 1st prize.

Areas of interest

CO₂ capture and valorisation

Hydrogen and synthetic fuels

Process design and simulation

Energy and process optimisation

Oil and gas

Base chemicals and polymers

Process intensification

Effluent treatment